

Part I: IQ Signal Modeling and Dataset Construction

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1 Introduction & Signal Modeling

In the first part of this project, a fully controlled dataset was constructed based on digital communication signals in the Complex Baseband. The generated signals are mathematically represented as complex numbers in the form of $x = I + jQ$, where I is the In-phase component and Q is the Quadrature component.

A total of 15 different modulation schemes were implemented (M-ASK, M-PSK, M-QAM, M-APSK, M-HQAM). A strict prerequisite for the subsequent training of the neural network was the energy normalization of the symbols. All constellations were normalized so that the average energy per symbol is strictly equal to one ($E_s = 1$). This ensures that the machine learning model will learn to identify the structural geometry of the constellations rather than relying on absolute transmission power.

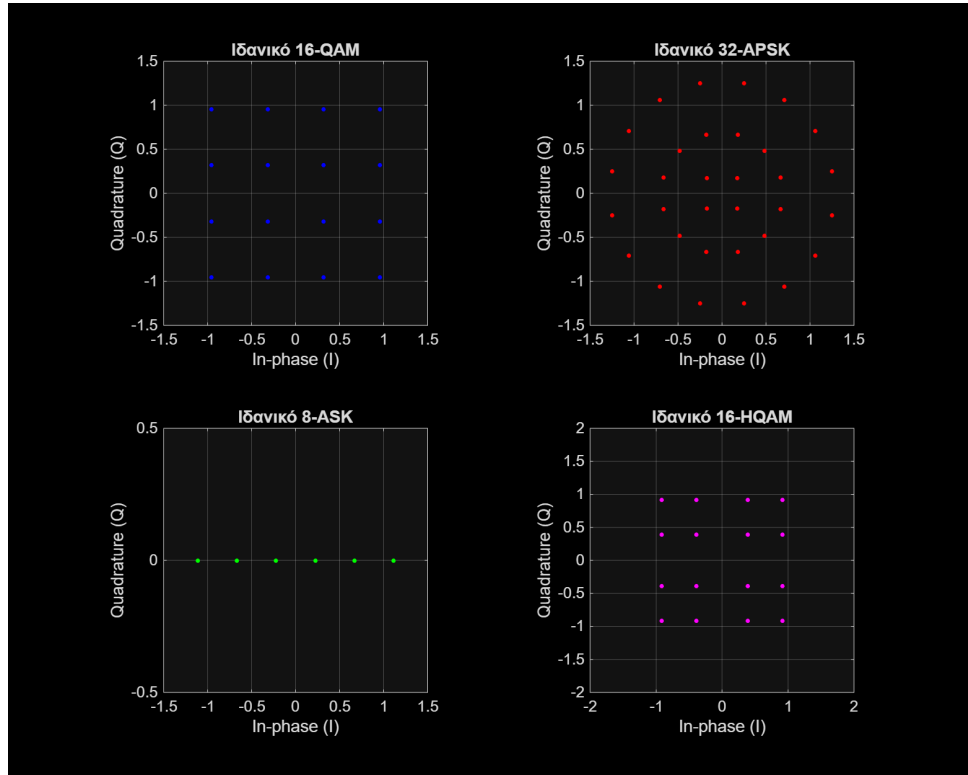


Figure 1: Ideal constellations of various modulation schemes.

2 Transceiver Impairments Modeling

To ensure the dataset accurately simulates realistic reception conditions, various impairments were introduced in the I/Q domain:

- **Phase Noise:** This was modeled as a multiplication of the signal by a complex exponential factor $e^{j\phi}$, where the phase ϕ follows a zero-mean Gaussian distribution, i.e., $\phi \sim \mathcal{N}(0, \sigma_\phi^2)$.
- **I/Q Imbalance:** The gain asymmetry (g) and phase asymmetry (θ) between the two branches of the local oscillator were simulated. The transformation was primarily applied to the Q component as follows:

$$Q_{imb} = (1 + g) (-I \sin(\theta) + Q \cos(\theta)) \quad (1)$$

- **Interference (Jamming):** An external deterministic interference was introduced in the form of a continuous wave (tone).
- **AWGN:** Additive White Gaussian Noise was added to the final signal, scaled appropriately to achieve the desired Signal-to-Noise Ratio (SNR).

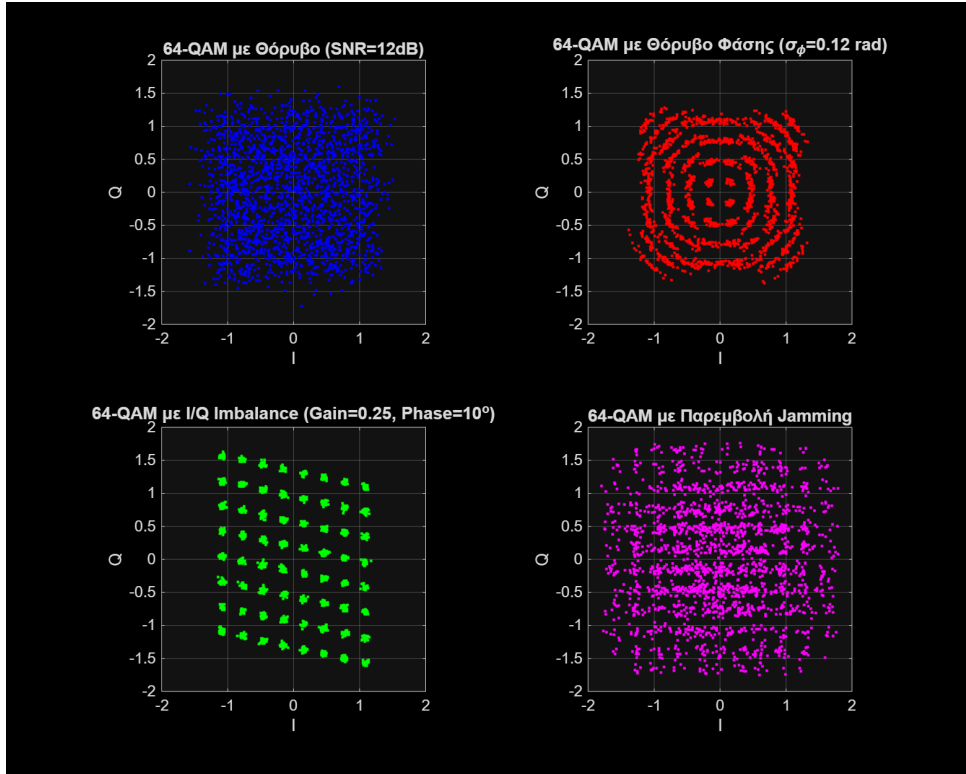


Figure 2: Effect of various impairments on a 64-QAM constellation.

3 Parameter Ranges & Justification

The simulation parameters were divided into discrete severity levels to generate categorical labels. The selection of these values is justified as follows:

Parameter	Values	Justification
SNR (dB)	5, 15, 25	5 dB (Low) represents high-error conditions, 15 dB (Medium) reflects typical operational conditions, and 25 dB (High) simulates a nearly ideal channel.
Phase Noise (σ_ϕ)	0, 0.08 rad	The value of 0.08 rad was chosen because it causes a distinct circular dispersion without completely destroying the topology of lower-order QAM schemes.
I/Q Gain Imbalance	0, 0.15	A 15% gain difference is considered a realistic hardware impairment in low-cost transceivers, noticeably distorting the constellation's squareness.

Table 1: Definition and justification of dataset parameters.

4 Dataset Construction & Label Structure

Each received signal was converted into a two-dimensional scatter plot. To make these images suitable for Convolutional Neural Network (CNN) training, the axes and labels were completely removed, and the plot limits were rigidly normalized to a fixed range of $[-2.5, 2.5]$. This preserves the spatial information strictly derived from the I/Q samples. Concurrently, a structured CSV file was generated, mapping the filename of each image to its predefined categorical labels (Modulation, Phase Noise, I/Q Imbalance, Interference, SNR Range).

5 SEP versus SNR Simulations

Finally, a Monte Carlo simulation was conducted to evaluate the Symbol Error Probability (SEP) as a function of the SNR for 16-QAM and 64-QAM constellations in the presence of phase noise.

As observed in the corresponding plot, the 64-QAM modulation is significantly more susceptible to errors. Furthermore, due to the presence of phase noise, the curves tend to exhibit an "error floor" at high SNR values, as errors are increasingly dominated by rotational phase distortion rather than thermal noise.

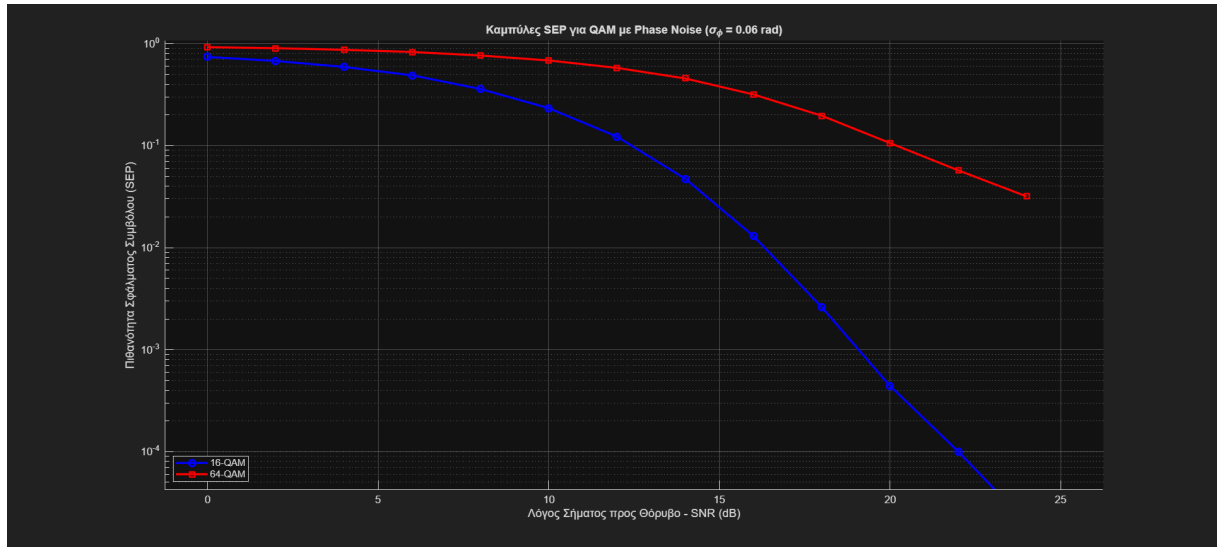


Figure 3: SEP versus SNR curves for QAM under the presence of Phase Noise.